



Balancing AI and academic integrity: what are the positions of academic publishers and universities?

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Abstract

This paper navigates the relationship between the growing influence of Artificial Intelligence (AI) and the foundational principles of academic integrity. It offers an in-depth analysis of how key academic stakeholders—publishers and universities—are crafting strategies and guidelines to integrate AI into the sphere of scholarly work. These efforts are not merely reactionary but are part of a broader initiative to harness AI's potential while maintaining ethical standards. The exploration reveals a diverse array of stances, reflecting the varied applications of AI in academic settings. These range from conservative approaches emphasizing the need for stringent AI regulations to more progressive strategies that seek to leverage AI's capabilities. The paper accentuates the importance of guidelines that are adaptable, transparent, and culturally sensitive, acknowledging the global context of academic research. It also highlights the pressing need for effective implementation and enforcement of these guidelines to ensure that the integration of AI into academia does not compromise the integrity of scholarly work. The conclusion emphasizes the dynamic balance required between embracing AI's transformative potential and upholding the timeless values of academic rigor and ethical research. This comprehensive analysis aims to provide a detailed perspective on the evolving role of AI in academia, fostering a dialog on how best to navigate this new technological frontier while preserving the sanctity of academic scholarship.

Keywords AI · Publishers · Academic institution · Research · Artificial intelligence · ChatGPT · LLMs

1 Introduction

In the dynamic interplay between technological innovation and academic rigor, the integration of Artificial Intelligence (AI) into scholarly pursuits presents a profound paradox (Malik et al. 2023). On the one hand, AI offers groundbreaking tools for data analysis, hypothesis generation, and even direct contributions to academic writing (Guersenzvaig and Sánchez-Monedero 2024; Bin-Nashwan et al. 2023). On the other hand, it poses fundamental challenges to the bedrock of academic integrity (Bin-Nashwan et al. 2023; Garcia and

Using 2024). As we explore the evolution of AI in academia, it's crucial to recognize the socio-technical landscape within which these technologies operate. The rise of Large Language Models (LLMs) and their integration into research and publishing have sparked a debate encompassing not just the technical capabilities of AI but also the ethical, regulatory, and societal dimensions of its application (Guersenzvaig and Sánchez-Monedero 2024; Munn et al. 2023; Toy 2023). This discussion extends beyond the functionality of AI tools to encompass broader concerns about decision-making processes, the ethics of automation in scholarly work, and the need for a framework that ensures transparency (Toy 2023), accountability, and trustworthiness in academic contributions (Garcia and Using 2024). This article endeavors to navigate this complex terrain, dissecting how major academic publishers and universities are formulating guidelines to harness AI's potential while safeguarding the sanctity of scholarly work.

The advent of AI in academic research and publishing is not a mere addition to the researcher's toolkit; it is a transformative force that redefines traditional methodologies

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and ethical paradigms (Bin-Nashwan et al. 2023; Abdelaal et al. 2019). As AI algorithms become increasingly capable of generating sophisticated, coherent, and, ostensibly, original research content, the academic community stands at a critical juncture. The quintessential values of authorship, originality, and methodological rigor are being recalibrated in light of AI's capabilities (Inam et al. 2024). This recalibration is not uniform; it is a mosaic of policies and perspectives, reflecting a diverse range of ethical, practical, and philosophical considerations. In navigating these waters, it becomes evident that the crux of integrating AI into academia lies not merely in its technological prowess but in addressing the underlying socio-technical systems (Johnson and Verdicchio 2024). These systems, which encompass the interplay between technology, people, and processes, are pivotal in shaping how AI is adopted and governed within academic settings. Through this lens, we examine the regulatory challenges and the collective responsibility of the academic community to foster an environment where AI is used responsibly—balancing innovation with integrity, and automation with accountability.

Major academic publishers, traditionally the gatekeepers of scholarly communication (Coser 1975), are at the forefront of shaping the discourse around AI in research. Their policies and guidelines on the use of AI tools in the publication process are pivotal in setting industry standards. These policies range from outright caution and stringent regulations to more open, albeit carefully monitored, acceptance of AI-assisted research. The nuances of these policies reflect the publishers' grappling with AI's implications for peer-review processes, citation practices, and the overall quality of published research. On the other hand, universities and research institutions, as the cradles of knowledge creation, are actively formulating their stances. Their guidelines not only impact how research is conducted within their precincts but also serve as a benchmark for academic conduct in the broader research community. These institutions are considering how AI intersects with academic integrity policies, particularly concerning data fabrication, plagiarism, and the ethical use of artificial intelligence in research (Faustine 2023).

The ethical debates surrounding AI in academia are far-reaching and multifaceted (Bin-Nashwan et al. 2023; Abdelaal et al. 2019; Inam et al. 2024). Central to these debates is the question of whether and how AI can coexist with the principles of academic integrity. Can AI-generated content be considered original? How do we ascribe authorship in works where AI plays a significant role? What are the implications for student learning and the development of critical thinking skills? These questions are not just academic; they have practical implications for the future of scholarly inquiry and the preservation of trust in academic outputs.

This article aims to provide a comprehensive analysis of the current landscape, examining the policies of major publishers and universities in response to the growing integration of AI in academic research. In undertaking this analysis, our intention is not to exhaustively catalog the positions of all entities within the academic sphere, such as the over 5000 publishers indexed in the Journal Citation Reports or the 1799 universities spanning 104 countries and regions as listed by Times Higher Education. Rather, the focus is selectively narrowed to a handful of leading and representative stakeholders. The randomly selected entities, therefore, serve as a representative sample, illustrating broader trends and challenges in AI integration across academia. Through this, we seek to illuminate the complexities of maintaining academic integrity in the age of AI, offering insights into how the academic community can navigate this new era without compromising the foundational values of scholarly work.

2 The rise of AI and the continuous debate over its application

In the scholarly field, the emergence and rapid advancement of AI has marked a pivotal transition, laden with both groundbreaking possibilities and significant ethical dilemmas (Bin-Nashwan et al. 2023; Abdelaal et al. 2019). As an illustration of this rapid advancement, a recent article in *Nature* (Ghassemi et al. 2023) points to a significant milestone of ChatGPT, a popular chatbot by the technology company OpenAI, designed to mimic human-like responses. This innovation quickly captivated both the scientific community and the public, amassing one million users in its first five days and now boasting over 180 million users (Ghassemi et al. 2023). The ascent of such technologies highlights the urgency and relevance of the current discourse on AI in academic settings. This discourse transcends mere technological adoption, touching upon the foundational aspects of academic integrity in the context of AI's growing role.

Tracing the trajectory of AI in academic spheres reveals its evolution from a tool for elementary data analysis to a sophisticated force capable of complex predictive analytics, natural language processing, and content generation (Chen et al. 2023). This dramatic evolution has initiated a redefinition of scholarly research, blurring the boundaries between human intellectual input and machine-generated contributions.

At the crux of the discourse is the role of AI in scholarly authorship and content generation. This development challenges the long-held perception of academic research as an exclusively human endeavor. It raises fundamental questions: Should AI be viewed simply as an auxiliary tool enhancing human intellect, or does it possess the capability

to contribute intellectually in its own right? Publishers and academic institutions are at the forefront of this conversation, crafting guidelines to address these novel challenges.

Leading publishers, for example, have adopted a stance that is cautiously optimistic about AI (CORE 2021). They acknowledge its potential to elevate research quality and streamline the publication process while maintaining strict guidelines to preserve the authenticity and integrity of scholarly work. These guidelines typically underline the necessity for transparency regarding AI's involvement in research, particularly in aspects like data analysis and interpretation. On the other side of the spectrum, there are reservations within the academic community regarding the rapid integration of AI into research (Graham 2022; Harris and Anthis 2021; Beckers 2018). Concerns range from the potential erosion of critical thinking and research skills in scholars to the likelihood of AI perpetuating existing biases in datasets. The capability of AI to generate content has also sparked apprehensions about the erosion of originality, raising red flags around issues of plagiarism and intellectual complacency (Bin-Nashwan et al. 2023; Abdelaal et al. 2019; Beckers 2018).

The ethical dimension of AI's utilization in academic research is of paramount importance. The core values of academic research—originality, transparency, and ethical conduct—are challenged by the introduction of AI.

3 Navigating AI's ethical, regulatory, and practical challenges in academia

In the complex world of academic research, the introduction of AI serves both as a source of innovation and a reason for ethical examination (Guersenzvaig and Sánchez-Monedero 2024; Bin-Nashwan et al. 2023; Garcia and Using 2024). As we stand at the confluence of technological advancement and scholarly tradition, the dialog between AI's potential and the venerable pillars of academic integrity unfolds, revealing a landscape rich with both promise and predicament (Toy 2023; Yeung et al. 2023). At the forefront of this exploration is the subtle interplay between AI technologies and the safeguarding of academic work (Bin-Nashwan et al. 2023; Toy 2023; McCradden et al. 2020). The dual capacity of AI to enhance scholarly pursuits and, conversely, to obscure the authenticity of intellectual contributions, calls for a vigilant recalibration of our ethical compass. As AI tools become increasingly woven into the fabric of research methodologies and publication processes (Garcia and Using 2024; Inam et al. 2024), the imperative to foster a domain where academic records remain unimpeachable grows ever more pressing. This segment of our journey highlights the necessity for robust ethical frameworks that can gracefully navigate the digital age's complexities, ensuring that AI serves

to illuminate rather than shadow the integrity of scholarly endeavors.

Further along the path, we encounter the labyrinthine challenges of bias and equity inherent in AI-driven predictive analytics and automated processes (Giovanola and Tiribelli 2023a, b). These tools, for all their efficiency and insight, carry the weight of potential biases (Giovanola and Tiribelli 2023a, b), casting long shadows over the fairness and integrity of research outcomes. Herein lies a call to arms for the development and implementation of equitable algorithms, and a continuous vigilance in the evaluation of AI tools, ensuring the research landscape remains a bastion of inclusivity and integrity.

As we chart this course, the significance of regulatory scaffolding and ethical codes in shaping the responsible application of AI tools within academic settings and beyond becomes abundantly clear (Bin-Nashwan et al. 2023; Toy 2023; McCradden et al. 2020). The regulatory landscape, far from being a static backdrop, is an ever-evolving framework, responsive to the cadence of technological innovation, ethical discourse, and the shifting contours of the academic research domain. In an effort to distill the essence of our discourse into a navigable blueprint, Table 1 encapsulates the key considerations laid forth.

Embarking on the journey of AI integration within the academic sphere demands more than just technological assimilation; it calls for a profound engagement with the ethical, regulatory, and practical dimensions that define this new frontier. In this evolving landscape, the role of guidelines and policies from publishers and universities is crucial. These entities, recognizing AI's transformative potential, are engaged in a delicate balancing act. Their policies are proactive efforts to shape the trajectory of AI in academic research. These evolving guidelines reflect the collective endeavor to leverage AI's capabilities while safeguarding the principles of academic integrity. Thus, the ongoing discourse over AI's application in academia mirrors a larger struggle to align technological advancements with the enduring values of academia. As AI continues to penetrate academic circles, this discourse is expected to evolve, necessitating continuous dialog and reassessment of the ethical and practical frameworks that govern its use. The path to balancing AI's potential with academic integrity's demands is intricate, necessitating the need for a vigilant and forward-thinking approach from the academic community.

4 Balancing AI advantages with scholarly ethics

In the dialog surrounding the incorporation of AI into the academic framework, a paramount consideration lies in harmonizing AI's technological advancements with the bedrock

Table 1 Challenges and potential implications for AI integration in academic integrity

Theme	Challenge	Potential implication	Source
Socio-Technical Systems	Integration of AI alters the interaction between technology and social structures	May lead to misalignment between technological advancements and social norms	Johnson and Verdicchio (2024), Farina et al. (2022)
LLMs	Large Language Models (LLMs) can generate content that blurs the lines of authorship	Threatens the integrity of scholarly work and authorship	Garcia and Using (2024), Chen et al. (2023)
Regulations	Existing regulations may not fully encompass the rapid advancements in AI	Insufficient governance could lead to unethical use of AI	Buiten (2019), Uddin et al. (2024)
Ethics	The ethical use of AI in academia necessitates clear guidelines	Without ethics, AI risks being misused, undermining trust	Garcia and Using (2024), Faustine (2023)
Trustworthiness	Ensuring AI-generated content and processes are trustworthy	Loss of trust in academic outputs if AI's role is not transparent	Bin-Nashwan et al. (2023)
Transparency	The opaque nature of some AI algorithms	Difficulty in understanding and trusting AI decisions	Toy (2023)
Responsibility	Assigning responsibility for AI-driven outcomes can be complex	Challenges in accountability for errors or ethical breaches	Toy (2023), CORE (2021)
Security	Protecting academic integrity and data in the face of AI integration	Potential for data breaches and academic fraud	Toy (2023), McCradden et al. (2020)
Accountability	Ensuring accountability in AI use within academic settings	Without accountability, ethical and operational risks increase	Toy (2023), Farina et al. (2022)
Prediction	AI's predictive capabilities may introduce biases	Skewed research outcomes and potential discrimination	Bin-Nashwan et al. (2023), Giovanola and Tiribelli (2023a)
Automation	The increasing automation of academic processes by AI	May reduce the human intellectual contribution to academia	Pisica et al. (2023)

of scholarly ethics. Within the circle of ethical academic practice and writing, the roles of both academic publishers and universities are indispensable and profoundly influential.

5 Analyzing publishers' use of AI guidelines

The landscape of scholarly publishing is undergoing a profound transformation, particularly with the integration of AI (Inam et al. 2024). A notable example of this change is the flexibility shown by the American Psychological Association (APA) in adapting their citation guidelines for AI models like ChatGPT (APA 2023). This flexibility indicates a broader shift in academic practices to accommodate the evolving role of AI. In an in-depth analysis of the guidelines provided by major publishers, Table 2 offers a comprehensive overview.

These guidelines, crafted by publishers, such as Elsevier, Springer Nature, Wiley, and others, commonly exclude AI from authorship, emphasizing the essential role of human intellect and responsibility in scholarly work. They also mandate transparent disclosure of AI's role in research, underscoring the importance of maintaining the integrity and authenticity of academic work. Moreover, these guidelines highlight the responsibility of authors to ensure the accuracy and originality of content generated with AI assistance. This reflects

a commitment to uphold scholarly standards, positioning AI as an augmentative tool rather than a replacement for human creativity and critical thinking. The guidelines also address ethical concerns about AI's potential to amplify biases present in data and the risk of undermining the essence of original scholarly work. The apprehension about AI's content generation capability, raising concerns about plagiarism and the erosion of creativity, reflects a broader philosophical debate on intellectual contribution in the digital age.

Overall, the publishers' AI guidelines reveal a landscape characterized by cautious optimism and ethical vigilance. While recognizing AI's potential to enhance research quality and efficiency, there's a clear consensus on the importance of balancing technological advancements with the preservation of academic integrity. This balance is indicative of the evolving nature of scholarly research in the digital era, emphasizing the indispensable role of human oversight in the responsible application of AI in academia.

6 University stances on AI and academic integrity

Universities worldwide are rapidly updating their ethical guidelines in response to the growing utilization and debates surrounding AI in academia. In the UK, for example, the

Table 2 AI in academic publishing: guidelines and policies of some selected publishers

Publisher	AI use stance and ethical considerations	Source
Committee on Publication Ethics	- AI tools do not qualify for authorship in research publications - Use of AI tools for text or image generation should be disclosed in the manuscript, specifically in the Acknowledgment section, with details of when and how the tools were used	CORE (2023)
Elsevier	- Allows the use of generative AI and AI-assisted technologies for improving language and readability - Requires appropriate disclosure as per the Guide for Authors	Elsevier (2024)
Springer Nature	- Committed to an ethically-focused approach to AI - Emphasizes responsible use of AI solutions, centering human values	Springer (2023)
Wiley	- AI-generated content (AIGC) cannot be considered for authorship - The use of AIGC must be transparently described; authors are responsible for content accuracy	Hames (2007)
Taylor & Francis	- Acknowledges the increased use of AI tools in academic research - AI tools must not be listed as an author but their use should be acknowledged	Robinson (2023)
Sage	- Recognizes the use of generative AI/LLMs and their impact on authorship assumptions and policies - Authors using generative AI must ensure factual correctness and appropriate referencing of their output - AI technology may be used but could be exploited for fabricating submissions or subverting peer-review	Sage (2024)
Oxford University Press	- Supports ICMJE definitions of authorship; AI tools do not qualify as authors - Use of AI must be disclosed in cover letters and manuscript sections	OUP (2024)
Cambridge University Press	- AI contributions in research content must be declared and clearly explained in publications - AI and Large Language Models (LLMs) do not meet Cambridge's authorship requirements due to accountability needs - Authors are responsible for the accuracy, integrity, and originality of their research, including any AI usage	CUP (2024)
Nature Portfolio	- LLMs like ChatGPT do not satisfy authorship criteria - AI use should be documented in the Methods or a suitable alternative part of the manuscript	Portfolio et al. (2024)
The Royal Society	- AI and AI-assisted technologies cannot be listed as authors - Authors must ensure work originality and familiarize themselves with publishing policies	Royal Society (2024)
American Chemical Society	- AI tools are not recognized as authors in academic publications - The use of AI tools for text or image generation must be clearly disclosed in the manuscript - Authors bear the responsibility for the accuracy and originality of content, even when AI tools are used - AI's role in research and writing should be transparent to maintain the integrity of the scholarly work	Banik et al. (2020)
Public Library of Science (Plos)	- Contributions by AI tools must be clearly reported - Authors are responsible for ensuring content accuracy and avoiding plagiarism	Plos (2024)
IEEE	- AI-generated content must be disclosed in acknowledgments - AI systems used for content generation should be identified	IEEE (2024)
The American Physical Society (APS)	- AI-based writing tools can be used for light editing but not for authorship - Use of AI tools should be disclosed to editors	APS (2023)
The American Psychological Association (APA)	- Authors must verify information provided by AI tools - AI tool use for specific purposes like editing must be disclosed	APA (2023)
Frontiers	- Generative AI technologies cannot be listed as co-authors - Authors must check the factual accuracy of AI-generated content	Frontiers (2024)
JAMA Network	- AI use for content creation or editing must be reported - Description of AI-generated content and tools used should be included	Flanagin et al. (2023)

Table 2 (continued)

Publisher	AI use stance and ethical considerations	Source
MDPI	• AI chatbots and LLMs do not meet authorship criteria • AI technology can be used in writing but must be declared	MDPI (2023)
American Association for the Advancement of Science (AAAS)	• Editor approval needed for AI-generated content, ensuring ethical use • AI-focused research exempted	Wong (2024)

esteemed Russell Group, encompassing top universities, such as Oxford and Cambridge, has developed principles that advocate for the ethical use of AI (Faustine 2023). These principles stress the need for AI literacy and the integration of AI into teaching and learning methodologies while maintaining academic integrity. In addition to that, the Oxford University explicitly states that AI tools, including advanced LLMs like ChatGPT, are not to be recognized as authors in research contributions due to their incapacity to assume responsibility for the content (Oxford University 2023). Similarly, Cambridge University has issued guidelines that consider the use of AI-generated content by students as academic misconduct, reflecting a stringent stance on maintaining academic standards (Rielly 2023).

In the United States, MIT's Schwarzman College of Computing and the MIT Washington Office are spearheading the creation of policy briefs focused on AI governance (MIT News Office 2021). This initiative aims to foster a technically informed discourse on governing AI in a manner that ensures safety and promotes the thriving of AI technologies. The AI Policy Forum, an assembly organized by MIT researchers, is set to present initial policy recommendations to manage the effects of AI and develop AI systems that align with societal values (MIT News Office 2021). This proactive approach by MIT exemplifies the ongoing efforts to address the complex implications of AI across various sectors, including finance, mobility, and healthcare. In a similar move, Stanford University has also addressed the challenges posed by AI tools, such as ChatGPT and DALL-E, in academic settings. Their policy equates the use of generative AI in coursework to receiving assistance from another individual, highlighting the necessity for clear disclosure in line with the Stanford Honor Code. The policy encourages instructors to establish transparent guidelines for AI tool use in their courses, adapting to the rapidly evolving capabilities of these technologies (OCS 2023).

Canadian universities are equally not left behind in setting these guidelines. The University of Melbourne, as an example, is advocating for a collaborative approach to AI governance in education, recommends involving the Australian Government in guiding ethical considerations and establishing communities of practice among universities, government bodies, and industry partners (The University of Melbourne 2023). This recommendation underlines the significance of

transparency in AI usage and the need for continued investment in AI research and development. In a similar approach, the University of Toronto in Canada offers preliminary guidance on the use of generative AI tools in graduate thesis work, emphasizing academic quality, research integrity, and transparency. The guidance outlines the requirements for AI tool use approval and documentation, highlighting the potential risks associated with these tools (UTSGS 2023).

In Asia, the National University of Singapore (NUS) has established comprehensive guidelines for the use of AI tools, underlining the importance of academic honesty (LibGuides 2024). These guidelines delineate proper and improper applications of AI in academia, emphasizing the users' responsibility to verify the accuracy of AI-generated information and maintain transparency in its usage (LibGuides 2024). The NUS guidelines serve as a framework for navigating the ethical complexities introduced by AI in educational environments. In Taiwan, National Tsing Hua University (NTHU) has introduced guidelines to enhance AI competencies in university education. The guidelines advocate for the responsible and transparent use of AI tools by teachers and students, emphasizing diverse applications of AI in academic settings (National Tsing Hua University 2023).

In Africa, the University of Cape Town and the American University in Cairo have released guides addressing the ethical use of AI in education. These guides focus on maintaining assessment integrity, effective teaching with AI, and ethical practices for students using AI tools (Helen 2023; AUC 2023). The guides reflect the institutions' commitment to navigating the challenges and opportunities presented by AI in the educational sector. While these broad stances capture the universities' philosophical approach to AI, the practical application of these principles in academic publishing is spearheaded by the institutions' research ethics committees.

7 The role of institutional ethics committees in addressing AI issues in publishing

Institutional ethics committees (IECs) play a pivotal role in maintaining the integrity and ethical standards of academic research, especially in the evolving landscape shaped by AI (Al-haimi et al. 2021; McKay et al. 2023). As AI

technologies become increasingly integrated into various aspects of academic publishing, the responsibilities and challenges faced by these bodies have grown both in complexity and importance. Traditionally, IECs have been tasked with safeguarding ethical research practices, ensuring the protection of human subjects, the responsible use of data, and the integrity of the scientific record (Al-haimi et al. 2021; Hine 2021). However, the advent of AI in academic publishing has necessitated a broadening of their purview. These committees are now confronted with novel ethical dilemmas that transcend conventional boundaries, such as determining the authorship of AI-generated content, ensuring the transparency of AI-assisted research, and addressing potential biases encoded within AI algorithms (Hine 2021; Lõuk 2023).

One of the primary issues that RECs face with the integration of AI into academic publishing is defining and attributing authorship. AI can significantly contribute to research outcomes, raising questions about the recognition of AI's role in scholarly publications. Committees through their various institutions have begun to establish guidelines that require clear disclosure of AI's involvement in research projects, ensuring that the contributions of AI tools are acknowledged without conferring authorship status (Al-haimi et al. 2021; Hine 2021), which traditionally implies responsibility and accountability for the content. Transparency in the use of AI technologies is another critical area of focus. In this regard, IECs are developing standards that necessitate researchers to disclose the extent of AI's involvement in their studies, including the algorithms or models used and the data on which these models were trained (McKay et al. 2023). This ensures that readers can assess the reliability of the findings and that the research can be replicated or scrutinized, preserving the integrity of the scientific record.

AI's potential to perpetuate or even amplify biases present in training data is a significant concern for IECs (McKay et al. 2023). Ethical guidelines are being formulated to address this challenge, requiring researchers to critically evaluate and report on the fairness of the AI tools employed in their work (Faustine 2023). This includes scrutinizing the datasets used for training AI models for representativeness and bias, a crucial step toward ensuring that research outcomes are equitable and just. Institutions have responded to these challenges by establishing or revising ethical guidelines specific to AI's use in research and publishing as we already seen. In addition, some universities through the IECs, now offer training sessions and resources aimed at educating researchers about the ethical considerations of using AI, including workshops on responsible AI use, seminars on AI ethics, and consultation services for AI-related research projects. IECs are also increasingly engaging with interdisciplinary experts to navigate the ethical landscape

of AI in publishing. By collaborating with computer scientists, ethicists, and legal scholars, these committees are better positioned to understand the capabilities and limitations of AI technologies and to develop ethical guidelines that reflect the complexity of these tools.

As AI continues to evolve, so too will the ethical challenges it presents to academic publishing. IECs and institutions must remain agile, updating their guidelines and practices in response to new developments. This might include adopting more dynamic ethical frameworks that can adapt to the rapid pace of technological change and fostering a culture of ethical awareness and responsibility among researchers.

8 Refined approach to ethical AI integration in academia

Success in AI integration is evident in institutions like MIT and Stanford, which have not only established policies for AI use in academic settings but also actively engaged in fostering a dialog around AI governance, promoting safety, transparency, and alignment with societal values. Their development of policy briefs and efforts to encourage transparent guidelines for AI tool use in courses are proactive steps toward responsible AI integration. Similarly, academic journals, such as Springer Nature, the Science family, and PLOS ONE, have crafted their own guidelines for AI-generated content, each implementing varying levels of restriction and disclosure requirements to maintain integrity in scholarly publishing. The lack of uniformity in these guidelines, however, presents a potential loophole. Their inadequacies were highlighted in a recent controversy where a paper published in *Frontiers in Cell and Developmental Biology*, using Midjourney to depict a rat's reproductive organs (now retracted), featured a cartoon rodent with comically oversized genitalia, annotated with gibberish (Wong 2024), and yet passed peer review. This incident, along with numerous other papers featuring questionable AI content, serves as clear evidence of significant gaps in current oversight practices.

These kinds of challenges associated with the AI integration in academia necessitate a shift from passive acknowledgment to active ethical stewardship. This crucial transition calls for a deep examination of and response to foundational ethical questions introduced by AI in scholarly work, such as discerning the nature of originality in AI-generated content, defining the parameters of authorship, and understanding AI's impact on learning and critical thinking. Originality in AI-generated content requires distinguishing between mere data processing and genuine creative contributions, urging a redefinition of originality to include AI's innovative use as a tool while ensuring the preservation of the intellectual essence that constitutes unique scholarly insight. So also, in

tackling the issue of authorship, it's clear that while current guidelines unequivocally state AI cannot be authors, there's a need for the academic community to establish criteria that respect the collaborative nature of AI-assisted research without compromising the integrity of scholarly attribution. Echoing the proactive steps by the APA, which has adapted its citation guidelines to include AI content similarly to how software is cited, we recognize AI as a significant research tool. This approach avoids attributing human-like agency or authorial credit to AI, thereby ensuring accountability remains with human authors. Similarly, genuine concerns about AI's impact on learning and critical thinking demand an ethical approach to AI's integration into educational practices (Chan 2023). This approach should prioritize complementarity over replacement, leveraging AI to bolster pedagogical strategies and create an environment that encourages rather than inhibits critical thinking skills.

To navigate the complexities of AI integration across all levels of academia, from students and teachers to institutions and publishers, culminating in international collaboration, a comprehensive approach is essential. At the student and teacher level, fostering an understanding of ethical AI use within educational practices is crucial. This involves developing curricula that not only integrate AI technologies but also critically engage with the ethical implications of their use, ensuring that AI enhances pedagogical strategies without compromising the development of critical thinking skills. Institutions, on the other hand, bear the responsibility of creating and enforcing robust ethical frameworks that are adaptable and evolve with AI advancements. These frameworks should emphasize transparency, bias mitigation, and the ethical use of data, drawing parallels with the successful implementation of plagiarism detection tools like Turnitin. Such measures will not only safeguard academic integrity but also promote a culture of accountability and inclusivity. Publishers play a pivotal role in maintaining the integrity of scholarly publishing by establishing clear guidelines for AI-generated content, emphasizing vigilance and enforcement rather than relying solely on written guidelines. The development of unified standards for the review and publication of AI-assisted research will mitigate the risks of misinformation and uphold the quality of academic discourse.

To unify these efforts and address the ethical challenges of AI at a global scale, the establishment of an international body dedicated to AI ethics in academia is proposed. This body would standardize ethical guidelines, facilitate policy development, and encourage collaboration among educators, researchers, and policymakers worldwide. Its functions would include monitoring AI integration practices, offering ethical guidance, and serving as a central hub for sharing best practices and innovations. Such a comprehensive strategy ensures that the academic community can leverage AI's potential responsibly, fostering an environment where

technological advancements and ethical standards coexist harmoniously.

9 Limitations of the analysis

The comprehensiveness of this analysis is tempered by several a number of limitations. Primarily, the geographical focus on the Global North, achieved through a random selection and availability of information, may not fully capture the complete picture of AI ethics and integration challenges prevalent in the Global South, introducing a potential bias. Moreover, given the rapid evolution of AI technology, our findings, constrained by the scope of randomly selected institutions and reliance on publicly accessible information, offer a temporal snapshot that risks obsolescence as new developments arise. The lack of direct engagement with the chosen institutions further limits the depth of our insights into their specific AI ethics and integrity strategies. These constraints highlight the imperative for continuous, expansive research that embraces a more globally inclusive and engaged approach to understanding AI's role in academia.

10 Conclusion

As we reflect on the advancements made by universities and academic publishers in harmonizing the dynamics of AI with the principles of academic integrity, it's evident that significant progress has been made. These institutions have been instrumental in crafting thoughtful policies that leverage AI's capabilities while steadfastly protecting the foundational ethos of academia. Yet, the recurring controversies surrounding AI in academia highlight an imperative need for a unified approach that enforces consistency, accountability, and ethical stewardship across the educational landscape. Our firm stance advocates for the indispensable role of global collaboration and ethical vigilance in steering AI's integration into academic spheres, aiming to cultivate a milieu where technological advancements and ethical principles seamlessly converge. This mission transcends mere formulation of guidelines, pivoting toward their strategic execution and comprehensive assimilation within the academic fabric. It mandates a cohesive strategy enriched by the active participation of a broad coalition of stakeholders—students, educators, and technologists—ensuring the dynamic evolution of ethical frameworks that position AI as a catalyst for scholarly enrichment, not a hindrance. Advancing a proactive and foresighted approach is critical to nurturing an academic ecosystem where technological innovation and ethical integrity coalesce, fostering a culture of innovative and responsible inquiry. The proposition to inaugurate a global entity focused on AI ethics epitomizes

a deliberate stride toward synchronizing ethical standards and spearheading international collaboration, solidifying AI's role in enhancing intellectual pursuit underpinned by robust ethical governance. This forward-looking approach is vital for fostering a scholarly landscape where AI not only coexists with but amplifies human creativity and ethical inquiry, ensuring the sustenance of a vibrant, ethically grounded academic future.

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